MIL-L-65A
12 February 1968
SUPERSEDING
JAN-L-65
September 30, 1944
AMENDMENT 1
19 March 1953

MILITARY SPECIFICATION

LEAD THIOCYANATE (SULPHOCYANATE)

1. SCOPE

1.1 This specification covers lead thiocyanate for use in the manufacture of priming mixtures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard, Testing
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of
PPP-D-723 - Drums, Fiber

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington, D.C. 20036.)

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90 - Rules and Regulations of the Transportation of Explosives and Other Dangerous Articles

(Application for Copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

Standard D1193-66 - Reagent Water

Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Color and appearance. The material shall be white or yellow in color and free from visible impurities when tested as specified in 4.5.1.

3.2 Form. The material shall consist of discrete crystals when tested as specified in 4.5.2.

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3.3 Granulation. One hundred percent shall pass through U. S. standard sieve No. 140, and 75 percent through U. S. standard sieve No. 325 when tested as specified in 4.5.3.

3.4 Chemical and physical properties. Chemical and physical properties shall be as indicated in table I.

Table I. Chemical and physical properties

Properties	Requirements	Test paragraph
Lead as lead thiocyanate, % min.	99.3	4.5.4
Thiocyanate as lead thiocyanate, % min.	99.3	4.5.5
Chloride, as lead chloride, % max.	0.20	4.5.6
Sodium, as sodium thiocyanate, % max.	0.20	4.5.7
Moisture, % max.	0.20	4.5.8
Insoluble matter, % max.	0.20	4.5.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.2 Lot. A lot shall consist of a maximum of 1,000 pounds, produced by one manufacturer, at one plant, from the same materials and under

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4.3 Sampling. Sampling shall be done as follows: By means of a thief of the cement-sampling type, remove approximately one-half pound of material from each of 10 percent of the containers comprising the lot. A minimum of two containers must be samples. Mix the portions thoroughly and quarter until a composite sample of about 1 pound is obtained. Place this sample in an airtight container and label this so as to show the name of the material, manufacturer, plant, purchase order and number of pounds in the lot. This shall be the official sample representative of the lot.

4.4 Inspection. Inspection shall be made at the point of delivery, unless otherwise specified in the contract or order.

4.5 Test methods. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.5.1 Color and appearance. Take four portions of about 0.5 mg. each from different places in the sample and spread them over an area of about 2 sq. cm. on a glass microscopic slide. Examine the material macroscopically for color and impurities under daylight illumination and microscopically for impurities and aggregates of crystals using a magnification of about 30 times and reflected light.

4.5.2 Form. Select four portions each of approximately 0.5 mg. from different areas of the sample and spread over an area of approximately 2 sq. cm. on a glass microscopic slide. Observe the crystalline shape of the material microscopically using a magnification of about 30 times and reflected light.

4.5.3 Granulation. Place a U. S. standard sieve No. 140 on top of a U. S. standard sieve No. 325 and attach a receiving pan to the latter. Place 100 gm. of the sample on the top sieve, cover, and shake for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amount of material retained on each sieve and calculate the percentage of material passing through each sieve.

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4.5.4 Lead. Transfer 0.5 gm. of the sample, weighed to 0.1 mg., to 400 ml. beaker. Add 25 ml. of water and 3 ml. of concentrated nitric acid, cover with a watch glass and warm carefully to dissolve. Add 10 ml. of 1 to 1 sulfuric acid and evaporate to fumes of sulfuric acid. Cool, add about 75 ml. of water and warm to dissolve the soluble salts. Add 75 ml. of 95 percent ethyl alcohol and allow to stand for 2 hours. Filter on a tared sintered porcelain crucible of fine porosity (or a Gooch crucible with an asbestos mat). Transfer and wash with 3 percent sulfuric acid and then wash with 50 percent alcohol. Dry in an oven at 105° to 110°C for 30 minutes, then ignite in a muffle furnace at 550° to 600°C for 15 minutes. Cool in a desiccator and weigh. Calculate the percent lead thiocyanate as follows:

$$\text{Percent lead thiocyanate} = \frac{106.6A}{W} - 1.16 \times \text{percent lead chloride}$$

where: A = weight of precipitate, gm.
W = weight of sample, gm.

4.5.5 Thiocyanate.

4.5.5.1 Reagents.

4.5.5.1.1 Standard 0.1 N. silver nitrate solution. Dry silver nitrate at 120°C for 2 hours, cool, dissolve 16.989 gm. in water, and dilute to 1 liter in a volumetric flask. This is a primary standard.

4.5.5.1.2 Standard 0.1 N potassium thiocyanate solution. Dissolve 0.72 gm. of KCNS in water and dilute to 1 liter. Standardize against the standard silver nitrate solution.

4.5.5.1.3 Ferric ammonium sulfate indicator solution. Prepare a saturated solution of ferric ammonium alum in chloride free water and filter.

4.5.5.1.4 One percent nitric acid, made from water-white nitric acid.

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4.5.5.2 Procedure. Transfer a 0.65 to 0.70 gm. sample, weighed to 0.1 mg. to a 400 ml. beaker. Add about 150 ml. of 1 percent nitric acid and warm very gently to dissolve. Cool to room temperature. Add 50.00 ml. of standard 0.1 N silver nitrate solution with a buret and then add 0.5 ml. of ferric ammonium sulfate indicator solution. Titrate with standard 0.1 N standard potassium thiocyanate solution to the first permanent reddish coloration. Calculate the percent lead thiocyanate as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ [(D \times E)-(A \times B)]}{C} - 2.0 \times \text{percent sodium thiocyanate (from 4.5.7)}$$

where: A = ml. of potassium thiocyanate solution
B = normality of potassium thiocyanate solution
C = weight of sample, gm.
D = ml. of silver nitrate solution
E = normality of silver nitrate solution

4.5.6 Chloride.

4.5.6.1 Reagents.

4.5.6.1.1 Standard 0.05 N silver nitrate solution. Prepare by diluting standard 0.1 N silver nitrate solution (see 4.5.5.1.1) with an equal volume of water in a volumetric flask.

4.5.6.1.2 Standard 0.05 N potassium thiocyanate solution. Prepare by diluting standard 0.1 N potassium thiocyanate solution (see 4.5.5.1.2) with an equal volume of water in a volumetric flask. This solution is standardized against the standard 0.05 N silver nitrate solution.

4.5.6.1.3 Ferric ammonium sulfate indicator solution. (See 4.5.5.1.3).

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4.5.6.2 Procedure. Transfer a 10 gm. sample to a 250 ml. beaker, add 50 ml. of water, and digest just below boiling for 2 hours. Cool to about 15°C and filter through a medium texture filter paper. Wash with 25 ml. of water. Discard the precipitate. To the filtrate, add 10 ml. of 10 percent copper sulphate solution and 1 ml. of pyridine. Stir the mixture vigorously for a minute and allow to stand for 2 hours. Cool below 10°C and filter through a sintered glass of medium porosity. Transfer as much of the pasty material as possible to the crucible, but do not wash. Add 5 ml. of 1 to 1 nitric acid and 0.5 ml. of ferric alum solution to the filtrate. Add an excess of standard 0.05 N silver nitrate (10 ml. should be enough). Add 1 ml. of nitrobenzene and agitate the mixture vigorously. Titrate with 0.05 N potassium thiocyanate to the point at which the copper sulphate solution assumes a brownish tinge. Calculate the percent chloride as lead chloride as follows:

$$\text{Percent lead chloride} = \frac{13.91 \left[(D \times E - A \times B)\right]}{C}$$

where: A = ml. of potassium thiocyanate solution
B = normality of potassium thiocyanate solution
C = weight of sample, gm.
D = ml. silver nitrate solution
E = normality of silver nitrate solution

4.5.7 Sodium.

4.5.7.1 Reagents.

4.5.7.1.1 Magnesium uranyl acetate solution. Weigh 85 gm. of crystallized uranyl acetate ($UO_2(C_2H_3O_2)_2 \cdot 2H_2O$) in a 1500 ml. beaker, add 60 gm. of glacial acetic acid, dilute to about 1000 ml. with water, heat to about 70°C and stir until dissolved (= solution A). Weigh 500 gm. of crystallized magnesium acetate ($Mg(C_2H_3O_2)_2 \cdot 4H_2O$) into a 1500 ml. beaker, add 60 gm. of glacial acetic acid, dilute to about 1000 ml. with water, heat to about 70°C, and stir until dissolved (= solution B). Mix solution A and solution B at about 70°C and cool to 20°C. Add a few milligrams of sodium chloride and let stand for 24 hours at 20°C. Filter through a glass crucible of medium porosity and store in a pyrex glass-stoppered bottle. If, on long standing, a precipitate forms from interaction with the glass, the reagent should be filtered before use.

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4.5.7.1.2 Wash solution. Mix 1 ml. of 1 percent sodium chloride solution with 25 ml. of magnesium uranyl acetate solution. Filter the precipitate on a sintered glass crucible of medium porosity and wash with 95 percent alcohol. Suspend the precipitate in 1 liter of 95 percent ethyl alcohol, let stand for 1 hour while shaking frequently, and filter through a sintered glass crucible of medium porosity. The filtrate constitutes the wash solution.

4.5.7.2 Procedure. Weigh a 5 gm. sample into a 400 ml. beaker. Place the beaker under a hood and add 100 ml. of water and 10 ml. of nitric acid. The reaction between the thiocyanate and nitric acid once started is extremely vigorous. After the reaction has subsided, evaporate to about 50 ml., filter through a medium texture filter paper into a 150 ml. beaker, and wash with a total of 50 ml. of water. Discard the precipitate and evaporate the filtrate nearly to dryness. Add 25 ml. of water, neutralize to litmus paper with ammonium hydroxide, and add 3 ml. excess ammonium hydroxide. Filter through a medium texture filter paper into a 150 ml. beaker and wash with a small amount of water. Discard the precipitate and evaporate the filtrate to a volume of 3 to 5 ml. Add 25 ml. of magnesium uranyl acetate solution and allow to stand for 1 hour at 20°C while stirring occasionally. Filter the precipitate ($2NaC_2H_3O_2 \cdot 2Mg(C_2H_3O_2)_2 \cdot 6UO_2(C_2H_3O_2)_2 \cdot 13H_2O$) on a tared sintered glass crucible of medium porosity and transfer and wash the precipitate with a total of 25 to 30 ml. of wash solution. Dry at 100° to 110°C for 1 hour, cool in a desiccator, and weigh. Calculate the percent sodium as sodium thiocyanate as follows:

$$\text{Percent sodium thiocyanate} = \frac{5.40A}{W}$$

where: A = weight of precipitate, gm.
W = weight of sample, gm.

4.5.8 Moisture. Spread out 10 gm. of the sample on a watch glass, weigh the watch glass plus the sample, place in a sulfuric acid desiccator overnight, and weigh again. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100A}{W}$$

where: A = loss in weight, gm.
W = weight of sample, gm.

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4.5.9 Insoluble matter. Weigh a 5 gm. sample into a 1 liter beaker, add 800 ml. of a 5 percent solution of acetic acid and stir until the lead thiocyanate is dissolved. Filter through a tared glass crucible of medium porosity and wash with 5 percent acetic acid and then with water. Dry at 100° to 110°C for 2 hours, cool in a desiccator, and weigh. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100A}{W}$$

where: A = weight of residue, gm.
W = weight of sample, gm.

5. PREPARATION FOR DELIVERY

5.1 Packing and packaging.

5.1.1 Level A. Lead thiocyanate shall be packaged in a heat-sealed polyethylene bag in accordance with PPP-C-301 inside a lined fiber drum conforming to PPP-D-723, type III, grade A. Each drum shall not exceed 200 pounds ned. (See 6.1).

5.1.2 Level B. Lead thiocyanate shall be packaged in a heat-sealed polyethylene bag in accordance with PPP-C-301 inside a lined fiber drum conforming to PPP-D-723, type II, grade A. Each drum shall not exceed 100 pounds net. (see 6.1).

5.1.3 Level C. Lead thiocyanate shall be packed in water tight shipping containers in accordance with PPP-C-301 for shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation. (See 6.1).

5.2 Marking. In addition to any special marking specified, shipping containers shall be marked in accordance with MIL-STD-129. Each container shall be marked on the upper two-thirds of one surface as follows:

Contains Lead Thiocyanate
Toxic if inhaled or ingested
Use only in well ventilated area
Fine particles form explosive mixtures with air
Products of combustion are highly toxic
Firefighting - Use water, wear self-contained breathing apparatus
Store in cool, dry place, away from oxidizing agents.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size and type of shipping container
(c) Weight of lead thiocyanate to be packed in a shipping containers
(d) Level of packaging and packing desired.

6.1.1 Criteria for use of proper level of packaging and packing shall be as follows:

For level A - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For level B - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or domestic locations for an indefinite period.

For level C - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Custodian:
Army - MU

Preparing activity:
Army - MU

Review activities:
Army - MU, MD

Project No. 6810-0774

User activities:
Army - MI

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes" in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Philadelphia, Pa. 19137

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